LIGHT METRICS

Supply Macy

Team: Supplykart



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WHY SHIFT?

High ROI

Sustainable Savings

High Demand

WHAT (FLEET DIVISION) ?

60%

Light Commercial Vehicle (LCV)

30%

Medium Commercial Vehicle (MCV)

10%

Heavy Commercial Vehicle (LCV)

HOW?

Year 1

Replacement of **LCVs**

Year 2

Replacement of **MCVs**

Year 3

Replacement of **HCVs**

LIGHT METRICS

MAXIMUM EFFICIENCY STRATEGY

- **LCVs: Small battery packs with frequent short trips, charging at depots.**
- **MCVs: Mixed charging model with DC fast charging**
- **HCVs: High-priority DC fast charging for long hauls, with overnight depot charging.**

SECURITY

- Advanced Driver Assistance Systems (ADAS)
- **Smart Route Planning for EVs & Driver Monitoring**
- **Data-Driven Personalized Training Programs**

FEATURES

Automated Time-of-Use (ToU) Schedulin
Set charging to auto-initiate when rates drop below ₹7/kWh

Dynamic priority based charging
Route specific profile builder
Battery health guardrail (80% charge only)

Use Solcast API to predict solar yields at depot locations
Dynamic Scheduling: V2G Activation

VISION

"To revolutionize e-commerce logistics through sustainable, technology-driven solutions setting new standards for efficiency, safety, and customer satisfaction."

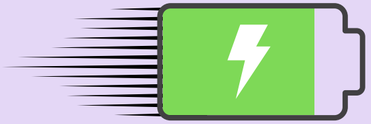
MISSION

Empowering India's e-commerce growth with efficient, sustainable, and customer-centric logistics solutions.

OPTIMIZING BATTERY HEALTH

Fast Charging

Deploy fast-charging stations in urban hubs



Slow (Level 2) Charging

Level 2 chargers in depots for overnight charging in suburban/rural areas

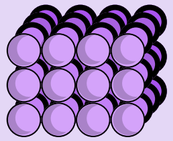


Use LFP batteries for short-haul/distribution trucks

HIGH CYCLE LIFE

NMC Batteries for long haul distances

MORE ENERGY DENSITY



BATTERY SWAPPING

For high-utilization routes, adopt battery-swapping stations to reduce downtime and deep discharges, as seen in China's success with 49.5% swap-capable trucks



REGENERATIVE BRAKING

Use of Regenerative Braking technology that recharges the battery on every brake. Extremely useful in city/ urban areas with traffic

Charging at Off-peak Hours

reduce costs and grid strain, particularly in regions with high time-of-use electricity rates

38%

Cost Reduction

18-24 Rs/kWh

Peak hour price range

6-8 Rs/kWh

Charging at Off-peak hours

2°C

Lower battery Temp

Auto Charge start when electricity rate drops to **7 Rs/kWh**

Dynamic Priority Based Charging	Route Specific Profile Building	Battery Health Guardrailing	V2G Activation (Future Prospect)

Use fleet management software to distribute charging loads, prioritize low-stress charging, and avoid simultaneous high-power draws



Maintaining the charge between 20 and 80% using the fleet management software

In hot climates (e.g., desert regions), prioritize trucks with advanced thermal management systems

In colder regions, precondition batteries during charging to maintain efficiency

TRAILER SPACE OPTIMIZATION

20%	Fewer trailers required using **CubeMaster 3D** load planning
15%	lower shipping costs
10%	CO_2 reduction via Cargo-Planner
28%	capacity boost with pallet pinwheeling techniques

Dynamic Truck Sizing & Fleet Mix

25%	Reduction in Delivery time

Pallets/Skids for Efficient Handling

20%	Damage Reduction
30%	Faster Loading and Unloading

ZF OptiLoad for Load Distribution

12%	Fuel Saving via axle weight optimization
15%	Lower Maintenance Cost
27%	Safer to transport in rough terrains

REDUCING EMPTY MILES

Route Optimization Software

Implement advanced AI-driven route planning tools to create efficient delivery routes, considering factors like traffic, delivery windows, and vehicle capacity. This can reduce empty miles by up to

64%
Reduction

Load Consolidation

Use Less-Than-Truckload (LTL) shipping to combine multiple smaller shipments into a single truckload, maximizing cargo space utilization and reducing empty return trips.

Backhaul Opportunities

Actively seek return cargo for otherwise empty trips, establishing partnerships with suppliers or offering return pick-up services to optimize vehicle usage.

Multi-Stop Routing:

Implement multi-drop deliveries, making multiple stops along a single route instead of point-to-point trips, ensuring trucks remain loaded throughout their journey.

Microtasking Platforms

Leverage platforms connecting individual couriers with smaller, local deliveries to optimize last-mile logistics and reduce urban empty miles.

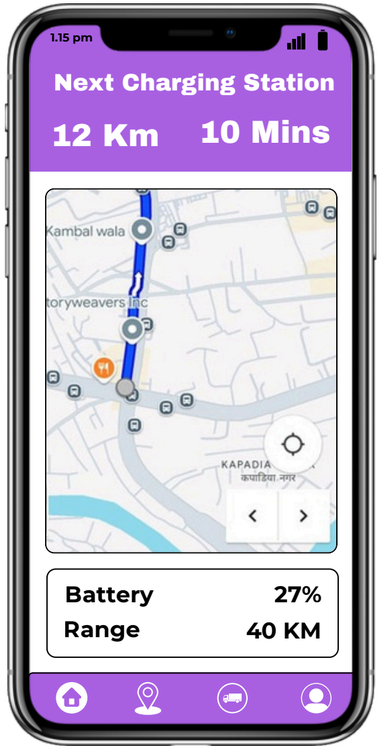
AI & Machine Learning

Predicts demand, optimizes routes, and automates workflows.

Fleet Management Software

Manages the operations of the entire fleet

- Safety Monitoring of the driver and vehicle
- Optimized Space Utilization and Load Consolidation
- Predictive Health Maintenance
- Optimized Route Planning
- Data driven consumer centric analytics
- Blockchain based chain of custody



Theft Prevention

1. Smart Locks and Access Control
- Implement IoT-enabled smart locks on trailers
- Use biometric authentication for drivers to access cargo areas
- Create a centralized access management system with real-time monitoring

2. GPS Tracking and Geofencing
- Install advanced GPS trackers on all vehicles and high-value cargo
- Set up geofencing alerts for unauthorized route deviations
- Implement real-time tracking accessible to both fleet managers and customers

3. Tamper-Evident Seals and Sensors
- Use electronic seals that log all opening/closing events
- Install motion and weight sensors in cargo areas to detect unauthorized movement
- Implement blockchain technology to create an immutable record of cargo integrity

KPIs

1. Fleet Electrification and Battery Optimization
- Percentage of fleet converted to EVs (target: 100%)
- Average battery health across the fleet (target: >90%)
- Charging efficiency (kWh consumed vs. miles driven)
- Percentage of charging done during off-peak hours (target: >70%)

2. Enhanced Freight and Vehicle Security
- Percentage reduction in theft incidents (target: >50%)
- Number of security breaches per 100,000 miles
- % of fleet equipped with advanced security systems (target: 100%)

3. Sustainability Metrics
- Total CO2 emissions reduction (compared to ICE fleet baseline)
- Energy consumption per ton-mile
- Percentage of energy sourced from renewables for charging

4. Financial Performance
- Total Cost of Ownership (TCO) for EVs vs. ICE vehicles
- Return on Investment (ROI) for EV transition
- Operational cost savings from electrification

Profile



Age: 35-55 Yrs
Background:
- Male
- Bachelor's in Operations
- 7+ years in fleet ops
- Manages 500 vehicles

Pain Points:
- Maintaining 98%+ EV uptime during ICE-to-EV transition
- Retrofitting maintenance bays for HV systems
- Driver resistance to telematics monitoring (67% adoption hurdles)

Psychology:
- Risk-averse decision maker needing TCO certainty
- Prioritizes operational continuity over innovation
- "Analysis paralysis" with conflicting KPIs: sustainability vs cost

Operations Manager

Profile



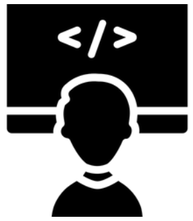
Age: 22-45 Yrs
Background:
- Male
- High school
- Avg 250 miles/week
- 12 hr shifts

Pain Points:
- Range anxiety in mixed terrain
- Unfamiliar EV regen braking patterns causing motion sickness
- Seat ergonomics worsen chronic back pain

Psychology:
- Resists telematics monitoring
- Values predictable routines over tech experimentation
- "Charging ritual" behaviours to combat anxiety

Truck Driver

Profile



Age: 28-40 Yrs
Background:
- Male / Female
- CS / EE Degree
- Manages 59+ IoT endpoints
- Cloud Certified

Pain Points:
- Legacy system-integration costs ($2.3M avg. for 500 EVs)
- Cybersecurity gaps in V2G communications
- API fragmentation across 8+ OEM platforms

Psychology:
- Solutions-oriented innovator frustrated by budget constraints
- PRefers open-source tools over vendor lock-in
- "Tech Whiplash" from rapid EV innovation cycles

IT Specialist

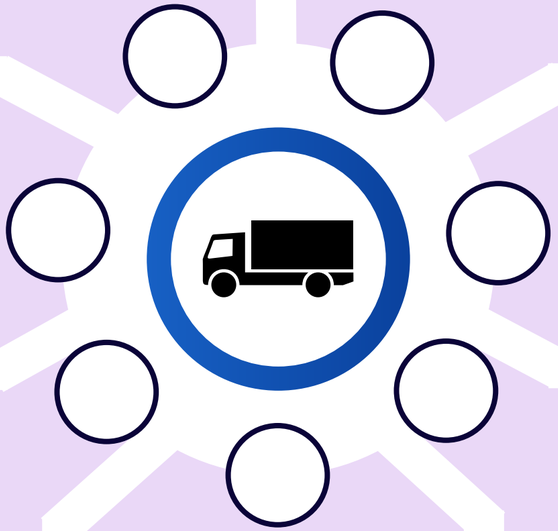
Introduction | Optimizing Battery Health | Space Optimization | Security | Cost Benefit Analysis

FINANCIALS

Capital Expenditure: 80.6 Crore

Drivers Total Training Costs: 9900000

OPERATIONAL SAVINGS: 2.16 Crore



Mantainence Savings: 1.35 Crore

Fuel Costs Savings: 74.23 Lakhs

ROI: 19.9%

Sustainibility Savings
Number of carbon credtis annually: 2000-3000

Total Cost of Charging = 17.68 lakhs

Overall distance travelled by fleet in a year = 2.2 Cr Kms

Number of charging stations required = 467

Appendix - 1

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Appendix - 2

Data

Conventional					Mileage	Distance	Litres of Diesel	Costs
Tata Ace Gold Petrol CX	175	443000	77525000	7047727.273	8	35000	4375	402500
Mahindra Bolero Pikup	75	1330000	99750000	9068181.818	8	35000	4375	402500
Eicher Pro 2049	50	1332000	66600000	6054545.455	6	50000	8333.333333	766666.6667
Tata 407 Gold SFC	75	1278000	95850000	37759090.91	6	50000	8333.333333	766666.6667
Eicher Pro 2049 CNG	50	1445000	72250000	28462121.21	6	50000	8333.333333	766666.6667
Tata 1512 LPT	25	2736000	68400000	26945454.55	6	50000	8333.333333	766666.6667
Tata Signa 4623.S	30	3650000	109500000	32075757.58	4	65000	16250	1495000
Ashok Leyland 4825 10x2 DTLA MAV	15	5052000	75780000	22198181.82	4	65000	16250	1495000
BharatBenz 4228R	5	5600000	28000000	8202020.202	4	65000	16250	1495000
Total			693655000	177813080.8				9192333.333

Electric					Battery Ca	Distance	Range	No of charges	Cost of Total charging
Tata Ace EV	175	1040000	182000000	182000000	21.3	35000	154	227.2727273	39937.5
Mahindra Atom EV	75	450000	33750000	33750000	11.1	35000	100	350	32051.25
Tata Ultra T.7 Electric	75	1849000	138675000	145913835	62.5	50000	110	454.5454545	234375
Eicher Pro 2055 EV	100	1591000	159100000	167405020	64.5	50000	162	308.6419753	164236.1
Ashok Leyland Boss 1218 HB EV	25	2882000	72050000	75811010	130	50000	350	142.8571429	153214.3
Ashok Leyland Boss 14 HB EV	30	4000000	120000000	132854980.8	201.5	65000	230	282.6086957	469801.6
PLTech Rhino 5536e	20	5000000	100000000	110712484	258	65000	205	317.0731707	674890.2
			805575000	848447329.8					1768506

Charging Infrastructure		
Every 150km		
Considering Highway Network of 70000 km (Excluding North East Kshmir and other states		
Overall No of Sttaions Requiired		466.666667
Per station		
Land	500000	
Elctrical Equipment	500000	
Chargers	494000	
Software	40000	
Team and other stuff	4,00,000	
Total	2127400	
Consider setting up 300 in 5 years	663748800	
Cosnidering the revenue from Charging Station		
Total Costs	848447329.8	

Carbon Credits saved Anually	1500	630000

Drivers	900	
Training	11000	9900000
Salary	400000	360000000
Insurance	20,000	18000000
Others	10000	9000000
Total		396900000

Fuel Costs	
Conventional	
Previous	9192333.33
Year 3	1768506.02
Year 1	7623655.42
Year 2	5108814.15
Savings	7423827.31

Annual Mantainence Costs	2%
Conventional	20809650
Electric	7250175
Savings	13559475
Year 1	15435150
Year 2	11842318.8

Implementation Roadmap

- **Month 1-3: Deploy tracking tools & customer portals**
- **Month 4-6: Train drivers/staff on new protocols**
- **Month 7-9: Roll out sustainability reporting**
- **Month 10-12: Optimize systems based on feedback**

Prevention of theft

1. Smart Locks and Access Control
- **Implement IoT-enabled smart locks on trailers**
- **Use biometric authentication for drivers to access cargo areas**
- **Create a centralized access management system with real-time monitoring**

2. GPS Tracking and Geofencing
- **Install advanced GPS trackers on all vehicles and high-value cargo**
- **Set up geofencing alerts for unauthorized route deviations**
- **Implement real-time tracking accessible to both fleet managers and customers**

3. AI-Powered Video Surveillance
- Deploy AI-enabled cameras inside and outside trailers
- Use computer vision to detect suspicious activities or unauthorized access
- Implement real-time alerts to security teams for immediate response

5. Secure Communication Systems
- Encrypt all data transmissions between vehicles, cargo, and control centres
- Implement multi-factor authentication for all digital systems
- Regularly update software to patch security vulnerabilities

6. Driver Training and Protocols
- Develop comprehensive security training programs for drivers
- Implement strict protocols for parking, rest stops, and cargo transfers
- Create a system for anonymous reporting of suspicious activities

7. Predictive Analytics for Risk Assessment
- Use AI to analyze historical data and predict high-risk routes or time periods
- Dynamically adjust security measures based on real-time risk assessments

Electric Vehicle-Specific Safety

Advanced BMS with Real time monitor

regenerative braking optimization

IoT Sensors Monitoring
(Energy Flows, Equipment Health, Consumption Patterns)

Cloud Analytics
(Centralized Data Platform & Edge Computing)

Predictive Analytics for Energy

Historical Data Analysis with Real Time Data

AI-Powered Optimization with Feedback Loops

Challenges

- Seasonal Variability
- Model Accuracy

Driver Specific

Advanced Driver Assistance Systems (ADAS)

- Autonomous Emergency Braking (AEB) with pedestrian detection
- 360-degree camera systems & Traffic jam assist
- Highway Pilot for semi-autonomous driving on expressways
- Predictive Cruise Control using topographical data

Conduct Audits | **Partner with Providers** | **Prioritize Routes** | **Train Drivers**

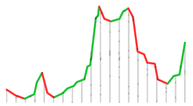
60% rear-end collisions

50% lane departure incidents

40% Overall accident

15% Insurance premium

Peer-to-Peer (P2P) Energy Trading and Dynamic Pricing



Energy Trading Platform



Customers with Excess Energy



Real Time Pricing



Blockchain Security

Challenges



Customer Trust & Transparency



Regulatory Framework for Market

Outcomes

15-20%
Reduction in Peak Load

30%
Greater Renewable Energy Usage

10-15%
Energy Savings for Households

FLEET ANALYSIS (500 TRUCKS)

Light Commercial Vehicle (LCV)

60%

Urban Delivery: 150
Last-Mile Delivery: 100
Small Parcel Carriers: 50

Medium Commercial Vehicles (MCV)

30%

Regional Transport: 75
Intercity Logistics: 50
Medium-Distance Haul: 25

Heavy Commercial Vehicles (HCVs)

10%

Long-Haul Transport: 30
Bulk Cargo: 15
Interstate Logistics: 5

Year 1 Replacement of Light Commercial Vehicle (LCV)

Year 2 Replacement of Medium Commercial Vehicles (MCV)

Year 3 Replacement of Heavy Commercial Vehicles (HCVs)

WHY SHIFT TO EV?

Sustainable Savings

High Demand

High ROI

OPTIMIZING BATTERY HEALTH



Maintaining the charge between 20 and 80%



Schedule Charge during cooler hours



Avoid Frequent fast charging

TEMPERATURE MANAGEMENT

Provide shaded or covered parking to protect vehicles from extreme temperatures

Use thermal management systems to regulate battery temperature during charging and operation

Precondition vehicle batteries before charging in cold weather

REGIONAL ADAPTATIONS FOR BATTERY HEALTH

In hot climates (e.g., desert regions), prioritize trucks with advanced thermal management systems and deploy charging stations with active cooling



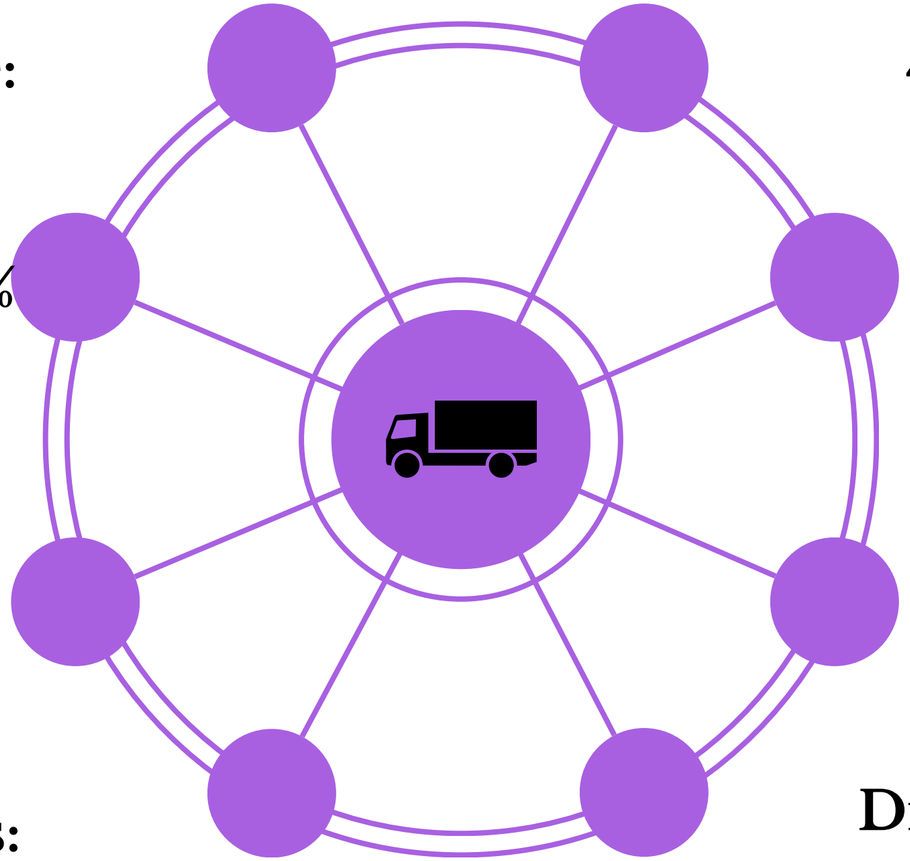
In colder regions, precondition batteries during charging to maintain efficiency

Capital Expenditure:
80.6 Crore

40% lower supply chain costs

ROI: 19.9%

Drivers:
900



OPERATIONAL SAVINGS: 2.16 Crore

Mantainence Savings:
1.35 Crore

Fuel Costs Savings:
74.23 Lakhs

Drivers Total Training Costs: 9900000